Muhammad Muizz Zafar

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Associate QA Engineer | Software Developer | Automation Testing

Technical Skills

Programming Languages:	C++, C#, Python, Java, JavaScript, TypeScript, GoLang, SQL, HTML/CSS
Frameworks/Tools:	NUnit, Google Test, Cypress, Jenkins, AWS, Docker, Vue.js, React, Angular, Git
Testing Expertise:	QA Automation, Test Case Design, Agile Testing, Regression Testing, Defect Tracking

Work Experience

Software Developer September 2024 - Present
Kinaxis Inc. Ottawa, ON

- Developed **C++ tooling** to automate migration of legacy **KUnit** test suites to the **Google Test (GTest)** framework, including serialization logic to transform test definitions within the Mocha codebase.
- Implemented and debugged **Scope Controller** functionality in C++, resolving feature incompatibilities between legacy and modular system architectures.
- Analyzed and debugged complex system behaviors in the supply chain planning engine, identifying root causes of discrepancies between the **Mocha** and **Latte** codebases.
- Designed automated validation queries and utilities to detect **known behavioral differences** across distributed modules, improving debugging and regression detection.
- Developed and maintained **GTest-based unit tests** and algorithm validation tests for newly implemented features in the planning system.
- Created a **C# test fixture** to orchestrate execution of C++ Google Tests and integrated it into **Jenkins CI/CD pipelines** using **Groovy**, improving automated regression testing and build reliability.

Full Stack Developer September 2023 - August 2024
University of Ottawa Ottawa, ON

- Enhanced the University's emergency alert website using **Laravel** and **Drupal**, improving accessibility and response speed for 20,000+ daily users.
- Designed and executed **unit and end-to-end tests** using **Jasmine**, **Cypress**, and **JUnit**, validating new frontend and backend features.
- Collaborated with engineering teams in an **Agile sprint cycle** to review test results, ensure feature reliability, and fix UI/API integration bugs.
- Optimized SQL queries and REST APIs in Java, achieving 15-second faster data lookups across the event system.

Cloud Developer Intern Sep 2022 - Dec 2022 & May 2023 - Aug 2023
Trend Micro Inc. Ottawa, ON

- Built automation for AWS-based testing environments using **GoLang**, **AWS Lambda**, and **Athena**, improving test data processing efficiency.
- Configured **CloudWatch** and **Splunk** alarms to proactively detect errors, reducing false test results and downtime.
- Developed and maintained automated integration tests for serverless pipelines to ensure stability of continuous releases.

Full Stack Developer Intern January 2022 - April 2022
Applied Mind Inc. Ottawa, ON

- Automated event detection workflows by integrating **Python**, **PostgreSQL**, and **GraphQL**, reducing manual QA verification time.
- Designed real-time dashboards in **Vue.js** for better monitoring and quicker validation of production events.

Full Stack Developer June 2019 - June 2021
Tech Under Twenty Oakville, ON

- Created and tested RESTful APIs using **.NET Core (C#)** for the member management system, improving stability and data validation accuracy.
- Designed responsive UI components in **Angular**, performing manual testing and debugging to improve user experience.

Education

University of Ottawa Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* September 2020 - March 2025

Projects

Delta Dispenser (Delta Hacks) *Dart, Flutter, Firebase* devpost.com/delta-dispenser

- Built a mobile app for medication tracking using Flutter and Firebase; won **1st place at DSC McMaster Challenge** among 300+ participants.